HEART DISEASE PREDICTION USING MACHINE LEARNING

Using Python, Pandas, NumPy, Matplotlib, Seaborn & Scikit-learn

Submitted By:
NIHAN EK

Objective & Goals

- **Objective**
- Develop a machine learning model to predict heart disease using patient medical records.
- **Goals**
- Load and analyze the dataset.
- Perform data preprocessing.
- Visualize important features.
- Train multiple machine learning models.
- Compare model performance.
- Predict heart disease accurately.

Introduction

About Heart Disease

- Heart disease is one of the leading causes of death worldwide.
- Early diagnosis can reduce mortality.
- Machine learning can assist doctors by predicting heart disease using patient information.
- Predictive models improve healthcare decision-making.

Dataset Information

- Use your output.
- Total Records : **1025**
- Features : **13**
- Target Variable : **Heart Disease**
- Total Columns : **14**

Dataset Features

Feature	Description
age	Age of patient
sex	Gender
cp	Chest Pain Type
trestbps	Resting Blood Pressure
chol	Cholesterol
fbs	Fasting Blood Sugar
restecg	ECG Results
thalach	Maximum Heart Rate
exang	Exercise Induced Angina
oldpeak	ST Depression
slope	ST Segment Slope
ca	Major Vessels
thal	Thalassemia
target	Heart Disease

First Five Rows

```
============================================================
FIRST FIVE ROWS
============================================================
   age  sex  cp  trestbps  chol  fbs  restecg  thalach  exang  oldpeak  slope  \
0   52    1   0       125   212    0        1      168      0      1.0      2
1   53    1   0       140   203    1        0      155      1      3.1      0
2   70    1   0       145   174    0        1      125      1      2.6      0
3   61    1   0       148   203    0        1      161      0      0.0      2
4   62    0   0       138   294    1        1      106      0      1.9      1

   ca  thal  target
0   2     3       0
1   0     3       0
2   0     3       0
3   1     3       0
4   3     2       0
```

Last Five Rows

```
     LAST FIVE ROWS
           age  sex  cp  trestbps  chol  fbs  restecg  thalach  exang  oldpeak  \
...  1020   59    1   1       140   221    0        1      164      1      0.0
     1021   60    1   0       125   258    0        0      141      1      2.8
     1022   47    1   0       110   275    0        0      118      1      1.0
     1023   50    0   0       110   254    0        0      159      0      0.0
     1024   54    1   0       120   188    0        1      113      0      1.4

           slope  ca  thal  target
     1020      2   0     2       1
     1021      1   1     3       0
     1022      1   1     2       0
     1023      2   0     2       1
     1024      1   1     3       0
```

Dataset Information

```
DATASET INFORMATION
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1025 entries, 0 to 1024
Data columns (total 14 columns):
 #   Column    Non-Null Count  Dtype  
---  ------    --------------  -----  
 0   age       1025 non-null   int64  
 1   sex       1025 non-null   int64  
 2   cp        1025 non-null   int64  
 3   trestbps  1025 non-null   int64  
 4   chol      1025 non-null   int64  
 5   fbs       1025 non-null   int64  
 6   restecg   1025 non-null   int64  
 7   thalach   1025 non-null   int64  
 8   exang     1025 non-null   int64  
 9   oldpeak   1025 non-null   float64
 10  slope     1025 non-null   int64  
 11  ca        1025 non-null   int64  
 12  thal      1025 non-null   int64  
 13  target    1025 non-null   int64  
dtypes: float64(1), int64(13)
memory usage: 112.2 KB
```

Overview

•The dataset contains **1025 patient records**.
•There are **14 columns**, including **13 input features** and **1 target variable**.
•All **1025 entries are non-null**, indicating there are **no missing values**.
•The dataset consists of **13 integer (int64) columns** and **1 floating-point (float64) column** (oldpeak).
•The total memory usage of the dataset is **112.2 KB**.
•The dataset is clean and suitable for preprocessing and machine learning model development.

Attribute	Value
Total Records	**1025**
Total Columns	**14**
Input Features	**13**
Target Variable	**1 (target)**
Missing Values	**0**
Data Types	**13 Integer, 1 Float**
Memory Usage	**112.2 KB**

Missing Values & Duplicate Records

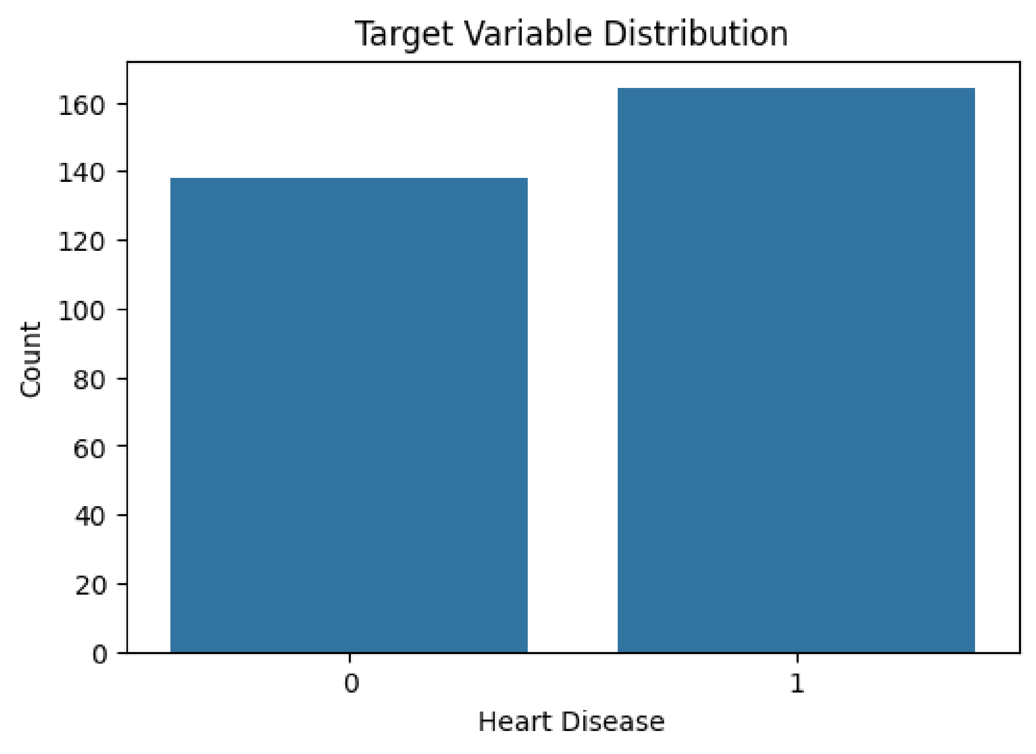
```
MISSING VALUES
age         0
sex         0
cp          0
trestbps    0
chol        0
fbs         0
restecg     0
thalach     0
exang       0
oldpeak     0
slope       0
ca          0
thal        0
target      0
dtype: int64

DUPLICATE RECORDS
723
```

Statistical Summary

```
STATISTICAL SUMMARY
              age         sex          cp    trestbps        chol         fbs  \
count  302.00000  302.000000  302.000000  302.000000  302.000000  302.000000
mean    54.42053    0.682119    0.963576  131.602649  246.500000    0.149007
std      9.04797    0.466426    1.032044   17.563394   51.753489    0.356686
min     29.00000    0.000000    0.000000   94.000000  126.000000    0.000000
25%     48.00000    0.000000    0.000000  120.000000  211.000000    0.000000
50%     55.50000    1.000000    1.000000  130.000000  240.500000    0.000000
75%     61.00000    1.000000    2.000000  140.000000  274.750000    0.000000
max     77.00000    1.000000    3.000000  200.000000  564.000000    1.000000
```

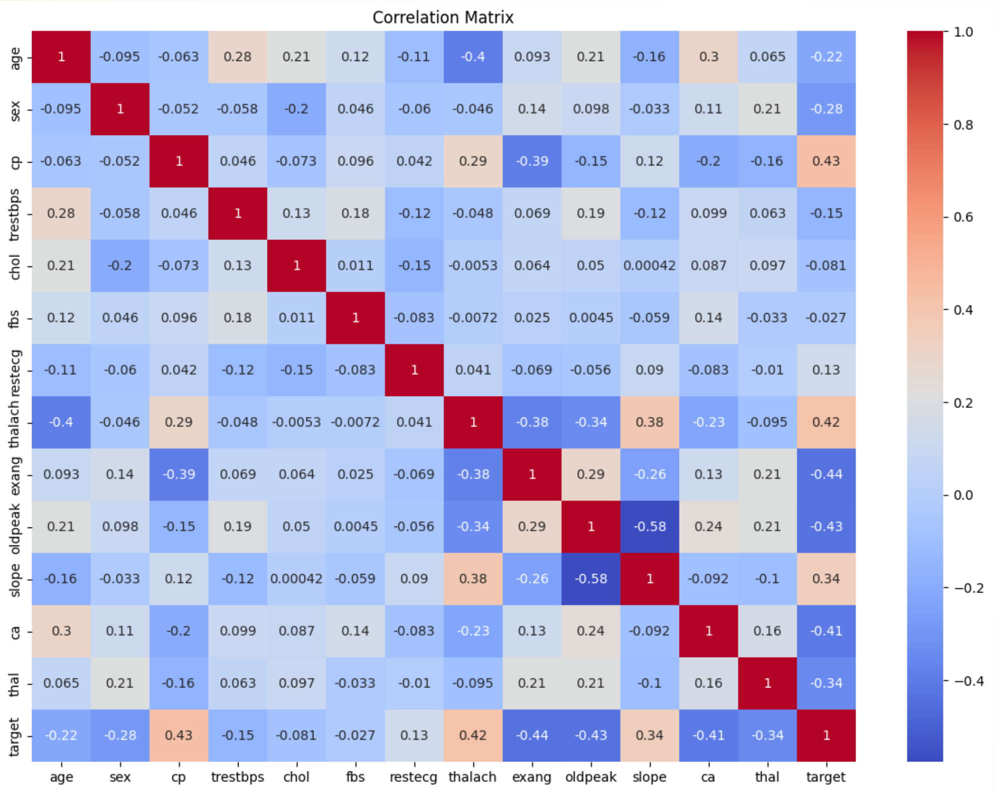
Parameter	Value
Total Unique Records	**302**
Average Age	**54.42 years**
Average Resting Blood Pressure	**131.60 mmHg**
Average Cholesterol	**246.50 mg/dL**
Average Maximum Heart Rate	**149.57 bpm**
Average Oldpeak	**1.04**
Heart Disease Cases	**54.30%**

Target Variable Distribution

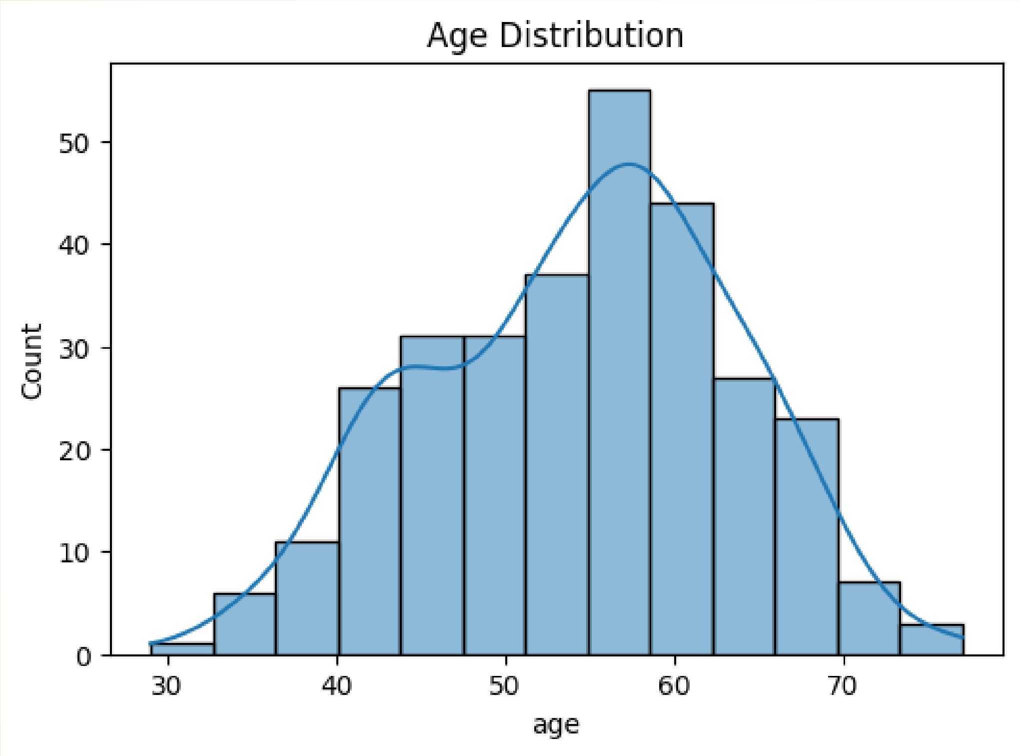


Target Class	Number of Patients	Percentage
Heart Disease (1)	**164**	**54.30%**
No Heart Disease (0)	**138**	**45.70%**

Correlation Matrix (Heatmap)



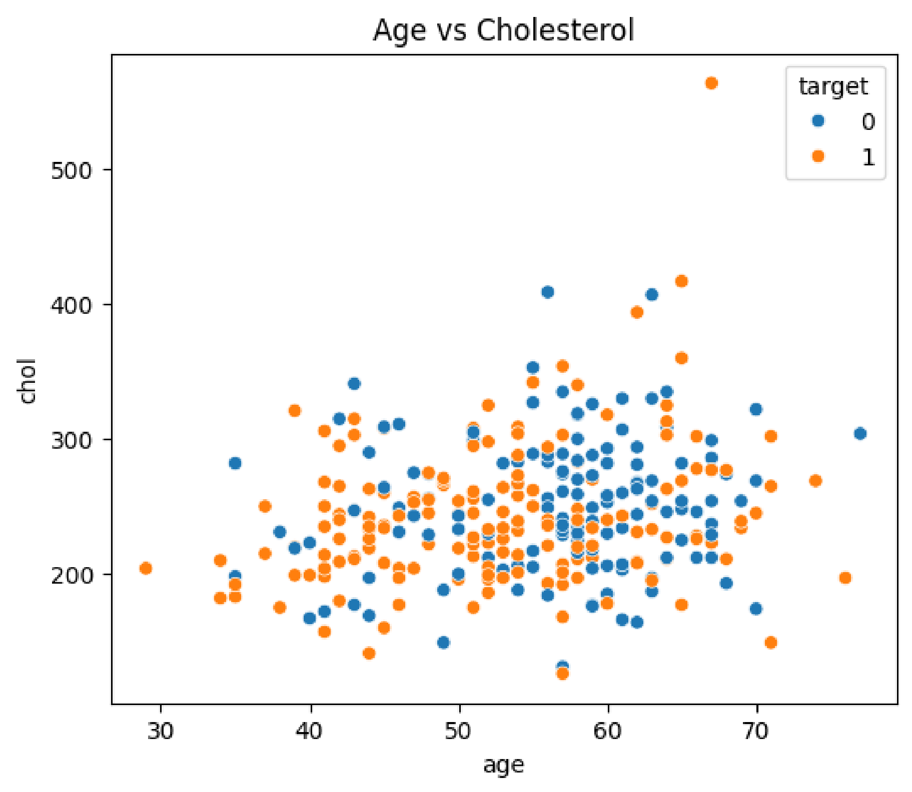
Age Distribution (Histogram)



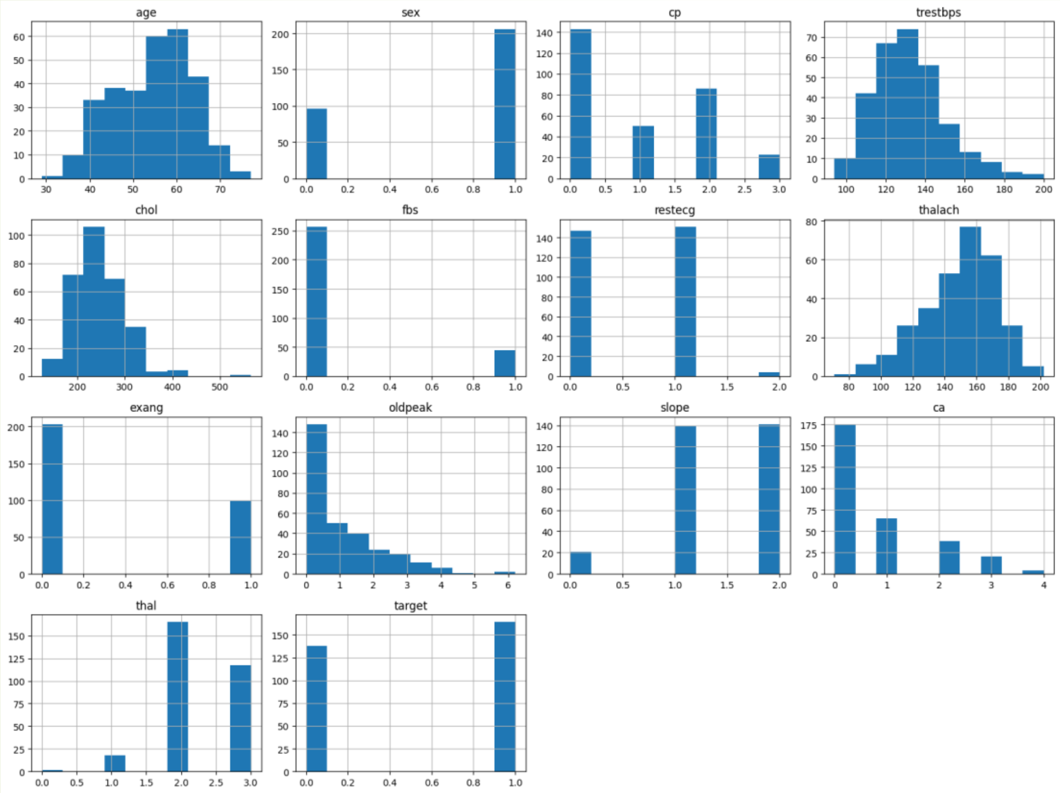
Summary Table

Parameter	Value
Minimum Age	**29 years**
Maximum Age	**77 years**
Average Age	**54.42 years**
Most Common Age Group	**45–65 years**

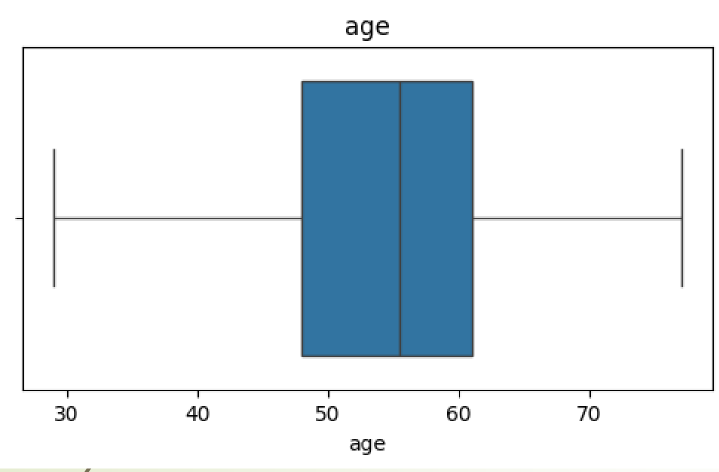
Age vs Cholesterol Analysis

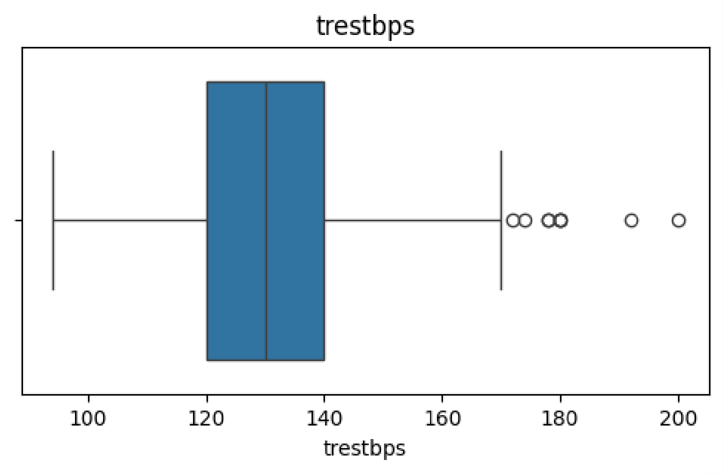


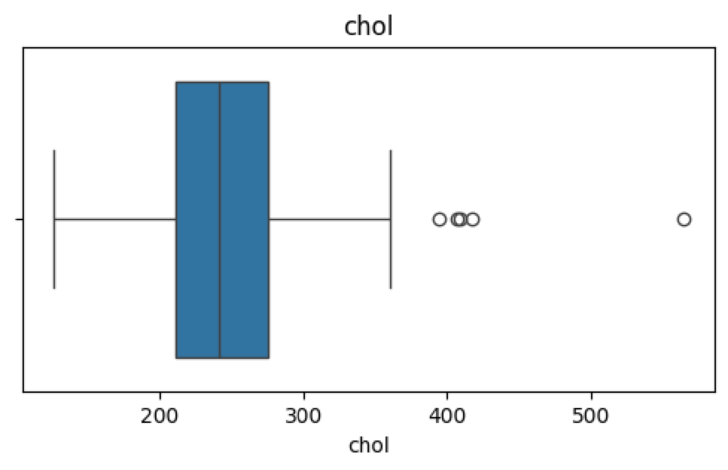
Pair Plot Analysis

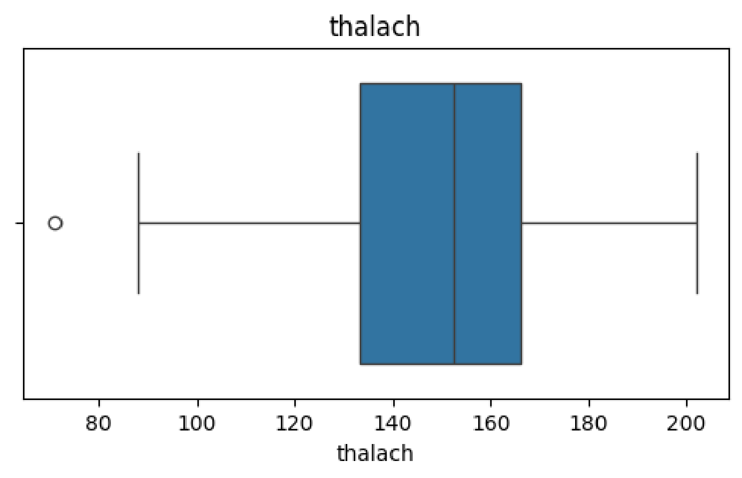


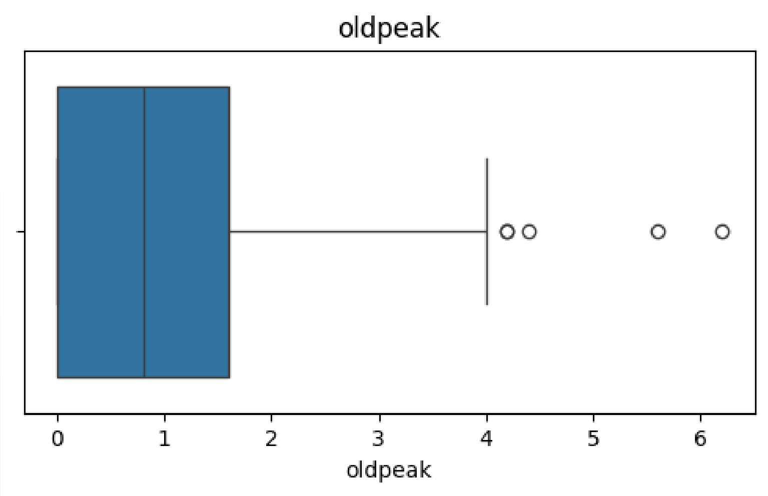
Box Plot Analysis











Data Preprocessing

Load Dataset

↓

Check Missing Values

↓

Remove Duplicate Records

↓

Feature Selection

↓

Train-Test Split (80% / 20%)

↓

Feature Scaling

↓

Machine Learning Models

Machine Learning Algorithms

Algorithms Used

1. Logistic Regression

•A statistical classification algorithm used for binary classification problems.

•Predicts the probability of heart disease based on input features.

•Simple, fast, and easy to interpret.

2. Decision Tree

Uses a tree-like structure to make classification decisions.

Splits the dataset based on feature values.

Easy to understand and visualize.

3. Random Forest

An ensemble learning algorithm consisting of multiple decision trees.

Produces more accurate and stable predictions than a single decision tree.

Reduces overfitting and improves generalization.

4. Support Vector Machine (SVM)
Finds the optimal decision boundary (hyperplane) to separate classes.
Performs well on high-dimensional datasets.
Effective for binary classification tasks.
5. K-Nearest Neighbors (KNN)
Classifies a patient based on the nearest neighboring data points.
Uses similarity (distance) to make predictions.
Simple and effective for smaller datasets.

Algorithms Comparison Table

Algorithm	Purpose
Logistic Regression	Binary Classification
Decision Tree	Rule-Based Classification
Random Forest	Ensemble Learning
Support Vector Machine	Hyperplane-Based Classification
K-Nearest Neighbors	Distance-Based Classification

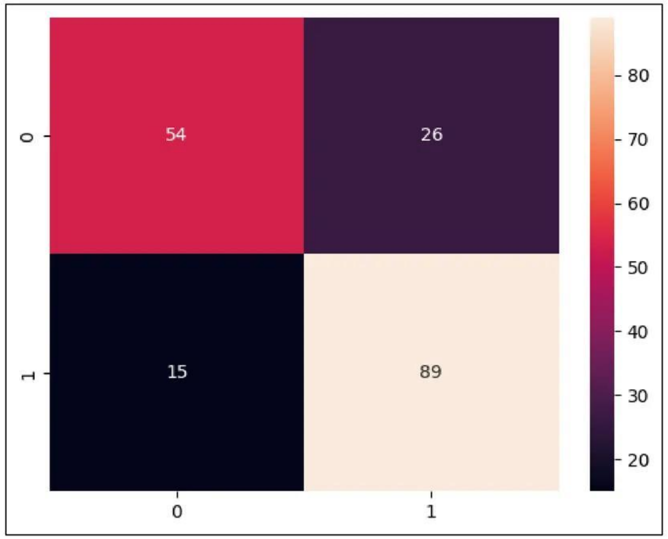
Logistic Regression Results

Model Evaluation

```
In [46]: from sklearn.metrics import accuracy_score
         print('The Accuracy Score is: ', accuracy_score(y_test,y_pred))

The Accuracy Score is:  0.7771739130434783
```

Precision, Recall, F1-Score, Support

```
In [47]: from sklearn.metrics import classification_report
         print(classification_report(y_test,y_pred))

              precision    recall  f1-score   support

           0       0.78      0.68      0.72        80
           1       0.77      0.86      0.81       104

    accuracy                           0.78       184
   macro avg       0.78      0.77      0.77       184
weighted avg       0.78      0.78      0.77       184
```



- The logistic regression has given an accuracy of **77.71**%.
- From the confusion matrix, we can say the model can classify whether the disease is present or not. But False Positives and False Negatives are also high to reduce this we will fit another classification model.

Model Performance

Accuracy	Precision (Class 1)	Recall (Class 1)	F1-Score (Class 1)
77.05%	0.70	0.90	0.79
Overall correctness	Positive prediction quality	Heart disease detection	Balanced performance

Decision Tree Results

```
===========================================================
DECISION TREE
===========================================================
Accuracy : 0.7377049180327869

Classification Report
              precision    recall  f1-score   support

           0       0.74      0.78      0.76        32
           1       0.74      0.69      0.71        29

    accuracy                           0.74        61
   macro avg       0.74      0.74      0.74        61
weighted avg       0.74      0.74      0.74        61
```

Model Performance

Metric	Value
Accuracy	**73.77%**
Precision (Class 1)	**0.74**
Recall (Class 1)	**0.69**
F1-Score (Class 1)	**0.71**

Random Forest Results

```
============================================================
RANDOM FOREST
============================================================
Accuracy : 0.8360655737704918

Classification Report
              precision    recall  f1-score   support

           0       0.89      0.78      0.83        32
           1       0.79      0.90      0.84        29

    accuracy                           0.84        61
   macro avg       0.84      0.84      0.84        61
weighted avg       0.84      0.84      0.84        61
```

Model Performance

Metric	Value
Accuracy	**83.61%**
Precision (Class 1)	**0.79**
Recall (Class 1)	**0.90**
F1-Score (Class 1)	**0.84**

Support Vector Machine (SVM) & K-Nearest Neighbors (KNN) Results

```
============================================================
SUPPORT VECTOR MACHINE
============================================================
Accuracy : 0.7868852459016393

Classification Report
              precision    recall  f1-score   support

           0       0.83      0.75      0.79        32
           1       0.75      0.83      0.79        29

    accuracy                           0.79        61
   macro avg       0.79      0.79      0.79        61
weighted avg       0.79      0.79      0.79        61
```

```
============================================================
K NEAREST NEIGHBORS
============================================================
Accuracy : 0.7377049180327869

Classification Report
              precision    recall  f1-score   support

           0       0.79      0.69      0.73        32
           1       0.70      0.79      0.74        29

    accuracy                           0.74        61
   macro avg       0.74      0.74      0.74        61
weighted avg       0.74      0.74      0.74        61
```

SVM Performance

Metric	Value
Accuracy	**78.69%**
Precision (Class 1)	**0.75**
Recall (Class 1)	**0.83**
F1-Score (Class 1)	**0.79**

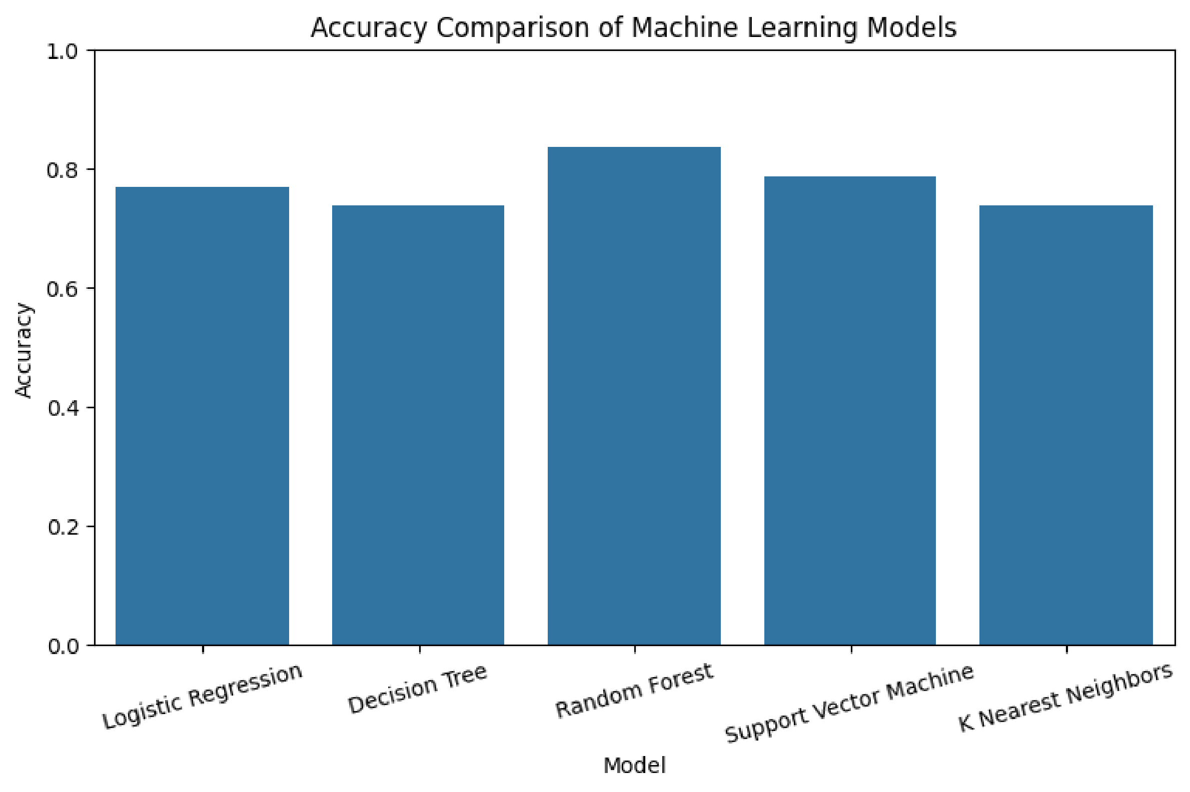
KNN Performance

Metric	Value
Accuracy	**73.77%**
Precision (Class 1)	**0.70**
Recall (Class 1)	**0.79**
F1-Score (Class 1)	**0.74**

Comparison

Model	Accuracy
Support Vector Machine	**78.69%**
K-Nearest Neighbors	**73.77%**

Model Performance Comparison

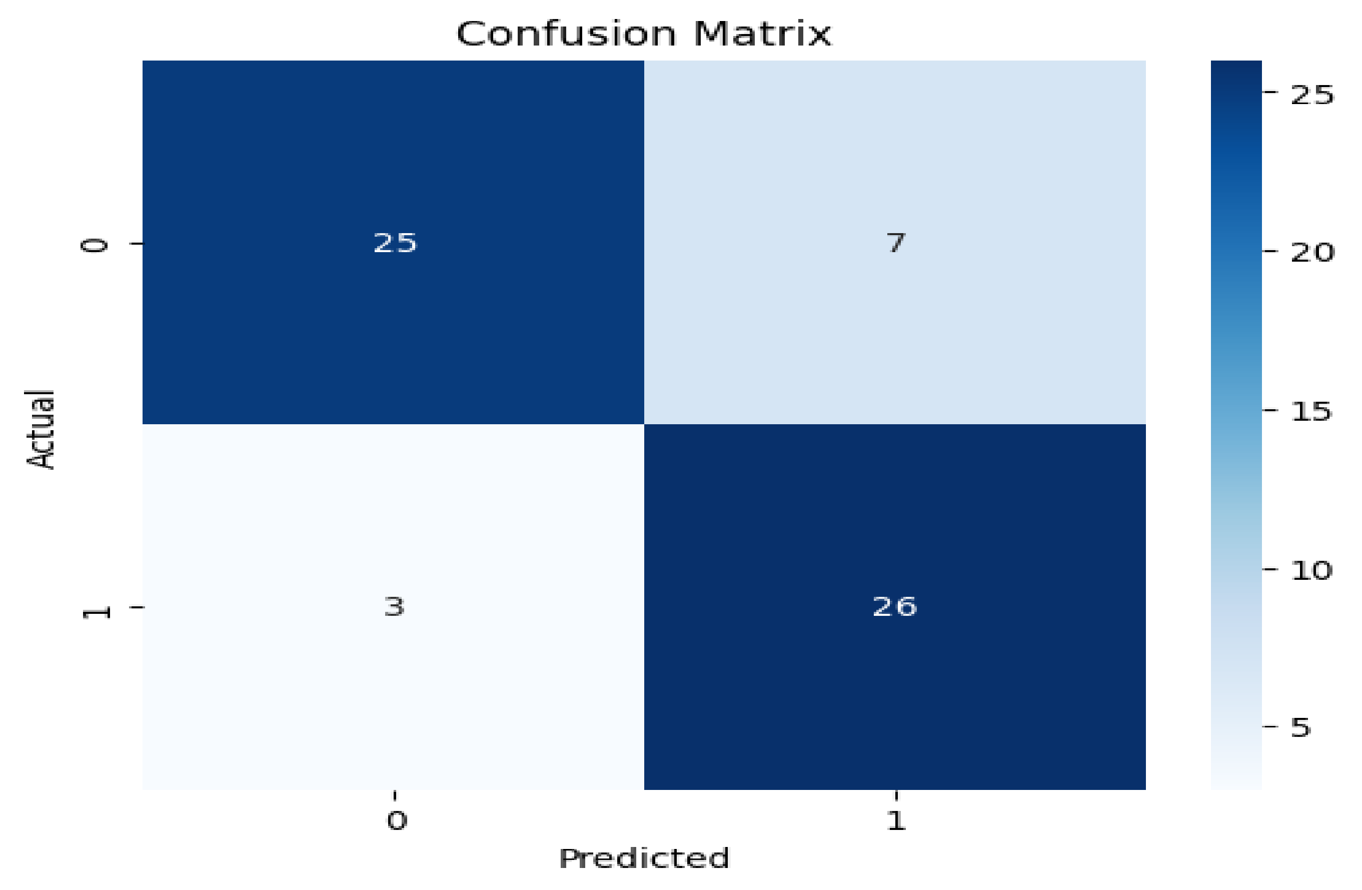


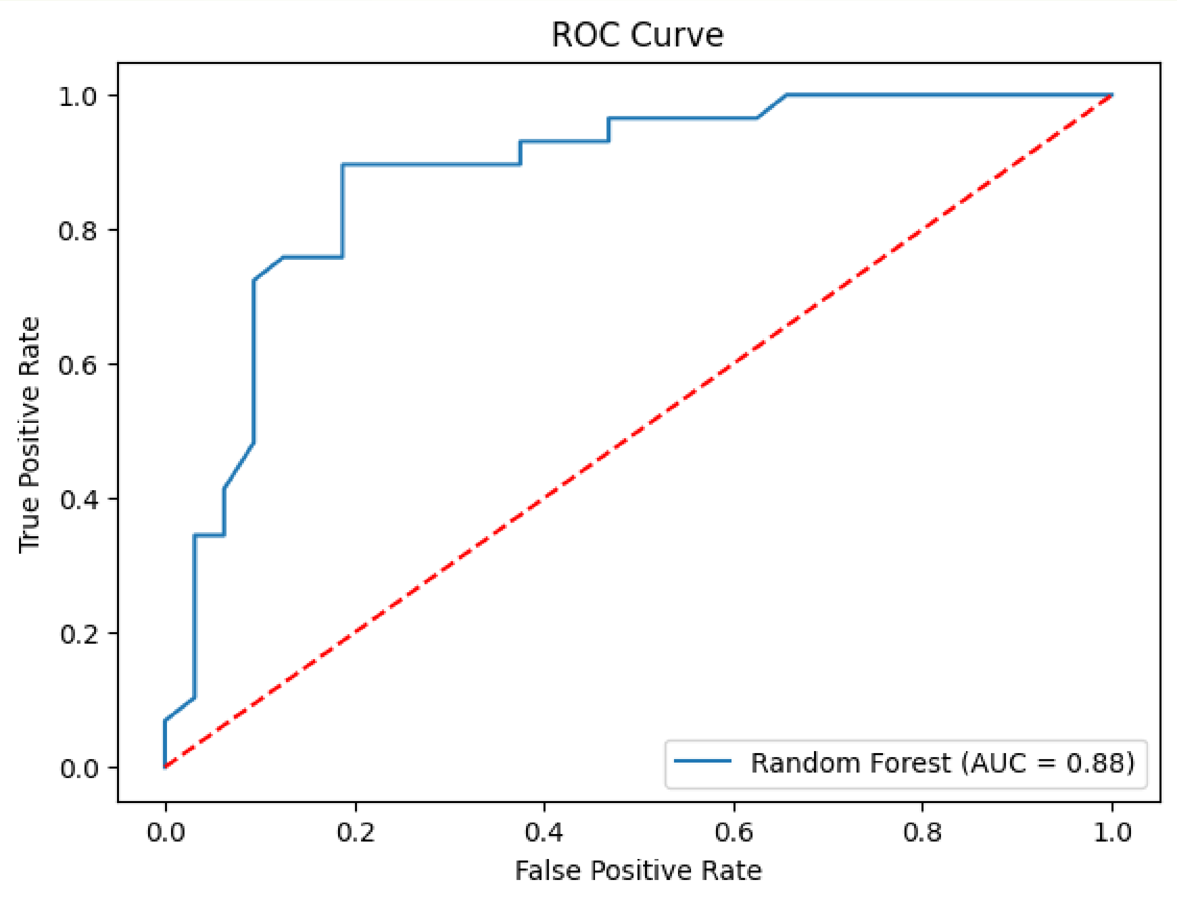
Model Performance Comparison

```
MODEL COMPARISON

                   Model  Accuracy
0    Logistic Regression  0.770492
1          Decision Tree  0.737705
2          Random Forest  0.836066
3 Support Vector Machine  0.786885
4    K Nearest Neighbors  0.737705

BEST MODEL

Model       Random Forest
Accuracy         0.836066
```

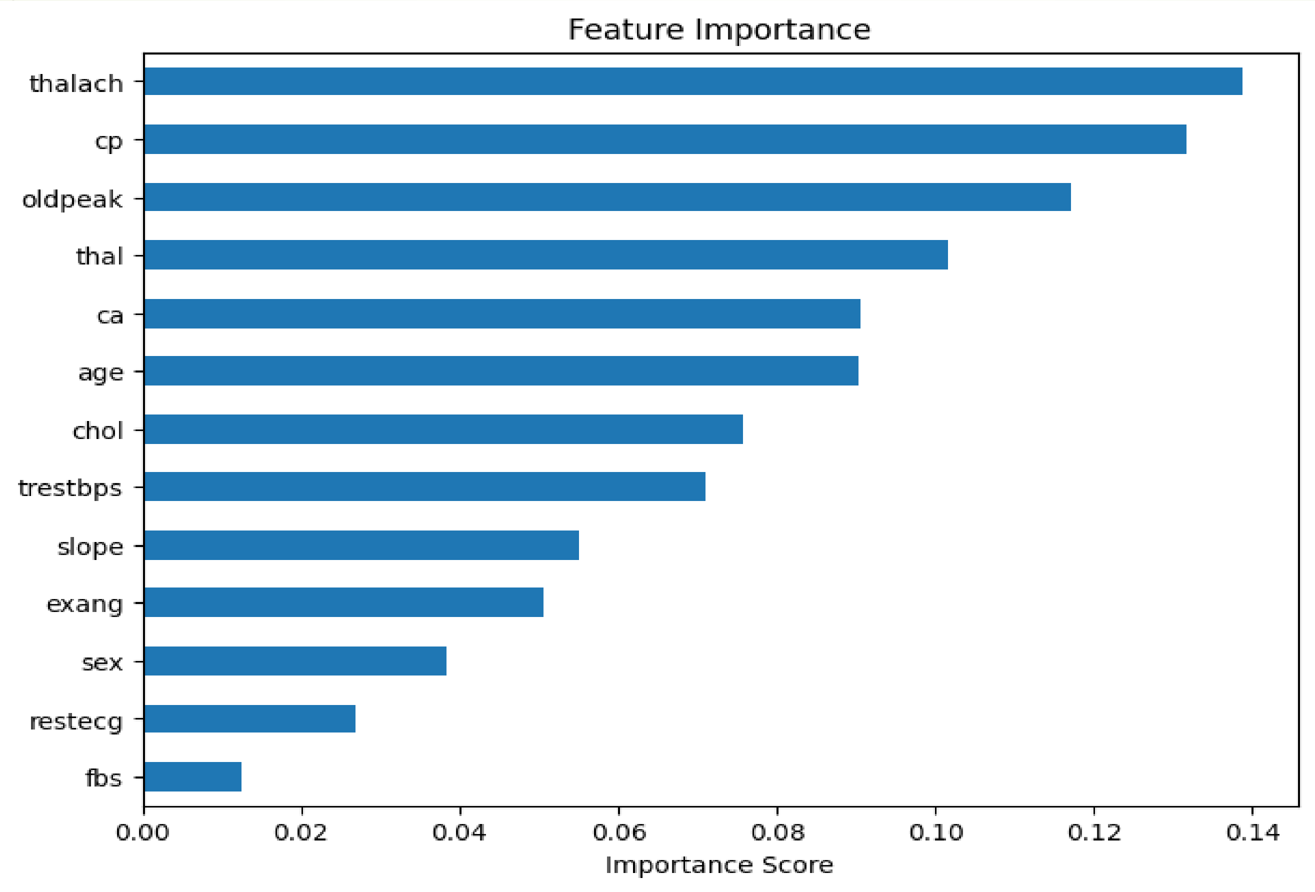
Algorithm	Accuracy (%)
Logistic Regression	**77.05%**
Decision Tree	**73.77%**
Random Forest	**83.61%**
Support Vector Machine	**78.69%**
K-Nearest Neighbors	**73.77%**

Model Evaluation – Confusion Matrix & ROC Curve




ROC Curve
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Random Forest (AUC = 0.88)

Feature Importance Analysis



Cross Validation and Model Saving

```
Cross Validation Scores
[0.83606557 0.81967213 0.9        0.8        0.78333333]
Average Accuracy
0.8278142076502732

Model Saved Successfully
Saved Model Loaded Successfully

Prediction : No Heart Disease
Probability : [[0.6 0.4]]
```

Cross Validation Results

Metric	Value
Number of Folds	**5**
Cross Validation Scores	**0.8361, 0.8197, 0.9000, 0.8000, 0.7833**
Average Accuracy	**82.78%**

Final Prediction and Project Summary

```
Prediction Result
No Heart Disease

============================================================
PROJECT SUMMARY
============================================================
Dataset Loaded Successfully
Data Preprocessing Completed
Multiple Machine Learning Models Trained
Model Evaluation Completed
Heart Disease Prediction Successful

THANK YOU
```

Final Prediction

The trained **Random Forest model** was used to predict the heart disease status of a sample patient.

Prediction Result

- **Prediction: No Heart Disease**
- The model classified the sample patient as **not having heart disease** based on the provided medical attributes.

Project Workflow

- Dataset Collection

  ↓

- Data Preprocessing

  ↓

- Exploratory Data Analysis

  ↓

- Feature Scaling

  ↓

- Train-Test Split

  ↓

- Machine Learning Models

  ↓

- Model Evaluation

  ↓

- Best Model Selection

  ↓

- Final Prediction

Key Findings

Key Findings

📊 Dataset Analysis

- The original dataset contained **1025 patient records**.
- **723 duplicate records** were identified and removed.
- The final dataset consisted of **302 unique patient records**.
- No missing values were found in the dataset.

Model Performance

Algorithm	Accuracy
Logistic Regression	**77.05%**
Decision Tree	**73.77%**
Random Forest	**83.61%**
Support Vector Machine	**78.69%**
K-Nearest Neighbors	**73.77%**

Conclusion

Project Conclusion

The **Heart Disease Prediction System** was successfully developed using machine learning techniques to classify patients based on their medical information. The project involved data preprocessing, exploratory data analysis, model training, evaluation, and final prediction.

Five supervised machine learning algorithms were implemented and compared to determine the most suitable model for heart disease prediction. Among them, the **Random Forest** algorithm achieved the highest accuracy of **83.61%**, making it the best-performing model for this dataset.

The project demonstrates that machine learning can effectively analyze patient health data and assist in the early detection of heart disease. Such predictive systems can support healthcare professionals by providing accurate and timely decision support.

Major Achievements
•Successfully loaded and analyzed the Heart Disease dataset.
•Removed duplicate records and prepared a clean dataset.
•Performed exploratory data analysis using graphs and statistical summaries.
•Implemented five machine learning algorithms.
•Compared model performance using evaluation metrics.
•Selected **Random Forest** as the best model with **83.61% accuracy**.
Successfully predicted heart disease using the trained model.

Significance in Healthcare
•Supports **early detection** of heart disease.
•Assists doctors in making **data-driven clinical decisions**.
•Helps identify **high-risk patients** quickly.
•Reduces diagnostic time through automated prediction.
•Demonstrates the practical application of **Artificial Intelligence in healthcare**.

Final Outcome
✓ Heart Disease Prediction System successfully developed.
✓ Machine learning models were trained and evaluated.
✓ Random Forest achieved the highest prediction accuracy.
✓ The system can assist healthcare professionals in improving patient diagnosis and treatment planning.

Future Scope

- **Future Enhancements**
- Although the developed Heart Disease Prediction System achieved good performance, several improvements can further enhance its accuracy, usability, and real-world application.
- **1. Use Larger and Diverse Datasets**
- Train the model using larger datasets collected from multiple hospitals.
- Improve the model's ability to generalize across different patient populations.
- **2. Improve Prediction Accuracy**
- Experiment with advanced machine learning and deep learning algorithms.
- Perform hyperparameter tuning and feature engineering to optimize performance.
- **3. Develop a Web or Mobile Application**
- Deploy the trained model as a web or mobile application.
- Allow doctors and patients to obtain predictions through an easy-to-use interface.
- **4. Real-Time Clinical Integration**
- Integrate the system with Electronic Health Record (EHR) systems.
- Enable real-time prediction using live patient data.

5. Include Additional Medical Parameters

Incorporate more clinical information such as:

- Body Mass Index (BMI)
- Smoking history
- Family history of heart disease
- Physical activity
- Blood sugar levels
- Lifestyle factors

6. Continuous Model Improvement

Periodically retrain the model using newly collected patient data.

Maintain high prediction accuracy as medical trends evolve.

Expected Benefits

-Improved prediction accuracy.

-Faster and more reliable diagnosis.

-Better support for healthcare professionals.

-Increased accessibility through web and mobile platforms.

-Enhanced patient care through early disease detection.

Thank You!

Heart Disease Prediction Using Machine Learning
Presented By : Nihan EK
Email : nihanekkarulai@gmail.com